What is Overleaf

Overleaf is an online editor used to write documents in LaTeX, a professional document preparation system widely used for:

- Research papers
- Project reports
- Thesis and dissertations
- Technical documentation
- Resumes/CVs
- Presentations
- Books and journals

The biggest advantage of Overleaf is that you don't need to install any software. You write your LaTeX code in the browser, click Recompile, and it automatically generates a PDF. It also supports collaboration, templates, version history, and bibliography management. (Overleaf)

Basic Structure of an Overleaf (LaTeX) Document:

```
\documentclass{article}
\begin{document}
Hello World!
\end{document}
```

Common LaTeX Tags (Commands):

1. Document Class:

```
\documentclass{article}
```

Other options:

```
\documentclass{report}
\documentclass{book}
\documentclass{letter}
```

2. Title Page:

```
\title{Student Performance Analysis}
\author{Rajaram Maremalla}
\date{\today}
\maketitle
```

3. Section Headings:

```
\section{Introduction}
\subsection{Background}
\subsubsection{Problem Statement}
```

4. Text Formatting:

Bold:

```
\textbf{Bold Text}
```

Italic:

```
\textit{Italic Text}
```

Underline:

```
\underline{Underlined Text}
```

5. Lists:

Bullet List:

```
\begin{itemize}
\item First Point
\item Second Point
\end{itemize}
```

Numbered List:

```
\begin{enumerate}
\item Step 1
\item Step 2
\end{enumerate}
```

6. New Line:

```
This is line one.\\
This is line two.
```

7. New Page:

```
\newpage
```

8. Insert Image:

```
\begin{figure}[h]
\centering
\includegraphics[width=0.8\textwidth]{image.png}
\caption{Sample Image}
\label{fig:image}
```

```
\end{figure}
```

9. Tables:

```
\begin{table}[h]
\centering
\begin{tabular}{|c|c|}
\hline
Name & Marks \\
\hline
Rajaram & 95 \\
\hline
\end{tabular}
\caption{Student Marks}
\end{table}
```

10. References:

```
\label{sec:intro}

See Section~\ref{sec:intro}
```

Important Packages Used in Most Reports

For internship reports, project reports, and assignments, these packages are commonly used:

```
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{graphicx}
\usepackage{xcolor}
\usepackage{hyperref}
\usepackage{fancyhdr}
\usepackage{titlesec}
\usepackage{setspace}
\usepackage{booktabs}
\usepackage{longtable}
\usepackage{array}
\usepackage{float}
\usepackage{caption}
\usepackage{subcaption}
\usepackage{parskip}
```

Packages You Use Frequently for Reports:

```
\documentclass[12pt,a4paper]{article}

\usepackage[a4paper,margin=1in]{geometry}
\usepackage{graphicx}
\usepackage{xcolor}
\usepackage{hyperref}
\usepackage{fancyhdr}
\usepackage{titlesec}
\usepackage{setspace}
\usepackage{float}
\usepackage{booktabs}
\usepackage{caption}
\usepackage{subcaption}
\usepackage{parskip}

\begin{document}

\title{Summer Internship Report}
\author{Rajaram Maremalla}
\date{\today}

\maketitle

\tableofcontents
\newpage

\section{Introduction}

\section{Weekly Progress}

\section{Dataset Analysis}

\section{Machine Learning Model}

\section{Conclusion}

\end{document}
```

Purpose of each Package

1. \usepackage[a4paper,margin=1in]{geometry}:

The Geometry package is used to control the layout of the document page. It allows the user to customize page dimensions, margins, header and footer spacing, and text area size. This package helps create professionally formatted documents by ensuring consistent page layouts according to institutional or publication requirements.

2. \usepackage{graphicx}:

The Graphicx package provides functionality for inserting and manipulating images within a document. It allows users to include figures, resize images, rotate graphics, and adjust their placement. This package is essential for reports containing charts, screenshots, diagrams, and other visual elements.

3. \usepackage{xcolor}:

The Xcolor package extends LaTeX's color capabilities. It enables the use of colors in text, tables, backgrounds, and other document elements. This package is useful for enhancing document readability and emphasizing important information through visual styling.

4. \usepackage{hyperref}:

The Hyperref package creates hyperlinks within the document and to external resources. It automatically generates clickable references for the table of contents, figures, tables, citations, and URLs. This improves document navigation, especially in digital PDF formats.

5. \usepackage{fancyhdr}:

The Fancyhdr package is used to customize page headers and footers. It allows users to display information such as document titles, chapter names, page numbers, author names, and dates at the top or bottom of each page. This package contributes to a professional document appearance.

6. \usepackage{titlesec}:

The Titlesec package provides advanced control over the formatting of section titles and headings. It allows customization of font size, style, spacing, numbering, and alignment of chapter, section, and subsection titles, helping maintain a consistent document structure.

7. \usepackage{setspace}:

The Setspace package is used to control line spacing throughout the document. It supports single, one-and-a-half, and double spacing formats. This package is particularly useful when preparing academic reports or theses that require specific spacing guidelines.

8. \usepackage{float}:

The Float package provides better control over the placement of figures and tables. It allows users to specify exact positions for floating objects, preventing them from being moved automatically by LaTeX. This ensures that visual elements appear close to their related content.

9. \usepackage{booktabs}:

The Booktabs package improves the quality and appearance of tables by providing professionally designed horizontal lines and spacing. It helps create cleaner, more readable tables suitable for academic and technical documents.

10. \usepackage{caption}:

The Caption package allows customization of captions for figures and tables. Users can modify the font style, alignment, numbering format, and spacing of captions, resulting in a more polished presentation of visual elements.

11. \usepackage{subcaption}:

The Subcaption package enables multiple images or tables to be grouped under a single figure or table environment. Each subfigure can have its own caption and label, making it easier to compare related visual elements within the same section.

12. \usepackage{parskip}:

The Parskip package improves document readability by adding vertical spacing between paragraphs and reducing the need for paragraph indentation. This creates a cleaner and more modern document layout.

13. \usepackage{bookmark}:

The Bookmark package enhances PDF navigation by creating organized and efficient bookmarks in the PDF viewer. It improves access to chapters, sections, and subsections, making large documents easier to navigate.

14. \usepackage{array}:

The Array package extends the capabilities of standard LaTeX tables by providing additional column formatting options. It allows greater control over alignment, spacing, and cell content within tables, making complex tables easier to design.

15. \usepackage{longtable}:

The Longtable package enables tables to span multiple pages automatically. This is particularly useful when presenting large datasets or extensive tabular information that cannot fit on a single page.

16. \usepackage[hidelinks]{hyperref}:

When the hidelinks option is used with the Hyperref package, hyperlinks remain functional while removing the colored borders and boxes that normally surround links. This results in a cleaner and more professional PDF appearance.